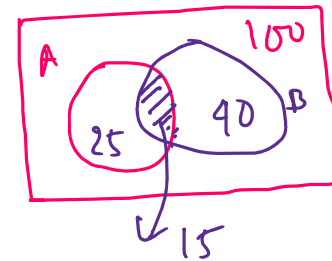


Event B $\rightarrow$ Being a DS

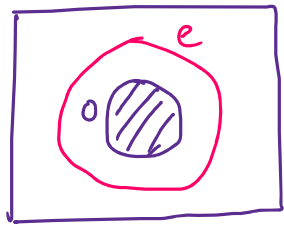
Event A $\rightarrow$ Being a python User.

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{15/100}{40/100} = \frac{15}{40} = \frac{3}{8}$$



$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$\Rightarrow \boxed{P(A \cap B) = P(B) \cdot P(A|B)}$$



$$P(A \cap B \cap C) = \frac{P(A \cap B \cap C)}{P(A \cap B)} \cdot \frac{P(A \cap B)}{P(A)} \cdot P(A)$$

$$= P(C|A \cap B) \cdot P(B|A) \cdot P(A)$$

$$= P(A|B \cap C) \cdot P(B|C) \cdot P(C)$$

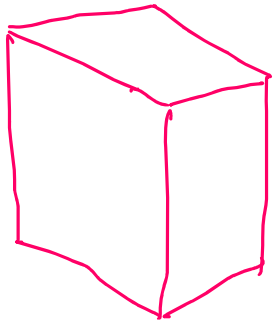
$$\begin{aligned}
 P(A_1 \cap A_2 \cap A_3) &= P(A_1) \cdot P(A_2|A_1) \cdot P(A_3|A_1 \cap A_2) \\
 &= \cancel{P(A_1)} \cdot \frac{P(\cancel{A_2} \cap A_1)}{\cancel{P(A_1)}} \cdot \frac{P(A_3 \cap A_1 \cap \cancel{A_2})}{\cancel{P(A_1 \cap A_2)}}
 \end{aligned}$$

$$P\left(\bigcap_{i=1}^n A_i\right) = P(A_1) \cdot P(A_2|A_1) \cdot P(A_3|A_1 \cap A_2) \cdot \dots \cdot P(A_n|A_1 \cap A_2 \cap \dots \cap A_{n-1})$$

my father goes to ____ .
 (w_1) (w_2) (w_3) (w_4) (w_5)
 (office)
 (market)
 (shop)

$$P(w_5 | w_1 \cap w_2 \cap w_3 \cap w_4)$$

next word prediction



Deck of 52
cards.

A_1 A_2 A_3 A_4 $\rightarrow$ 4 Aces.

first Draw

$$P(A_{CE_1}) = \frac{4}{52}$$

2nd draw

$$P(A_{CE_2} | A_{CE_1}) = \frac{3}{51}$$

3rd draw

$$P(A_{CE_3} | A_{CE_1} \cap A_{CE_2}) = \frac{2}{50}$$

$$\begin{aligned} \underline{P(A_{CE_1} \cap A_{CE_2} \cap A_{CE_3})} &= P(A_{CE_1}) \times P(A_{CE_2} | A_{CE_1}) \times P(A_{CE_3} | A_{CE_1} \cap A_{CE_2}) \\ &= \frac{4}{52} \times \frac{3}{51} \times \frac{2}{50} \end{aligned}$$

Independent Event

$$P(A|B) = P(A)$$

$$\Rightarrow \frac{P(A \cap B)}{P(B)} = P(A) \Rightarrow$$

$$P(A \cap B) = P(A) \cdot P(B)$$

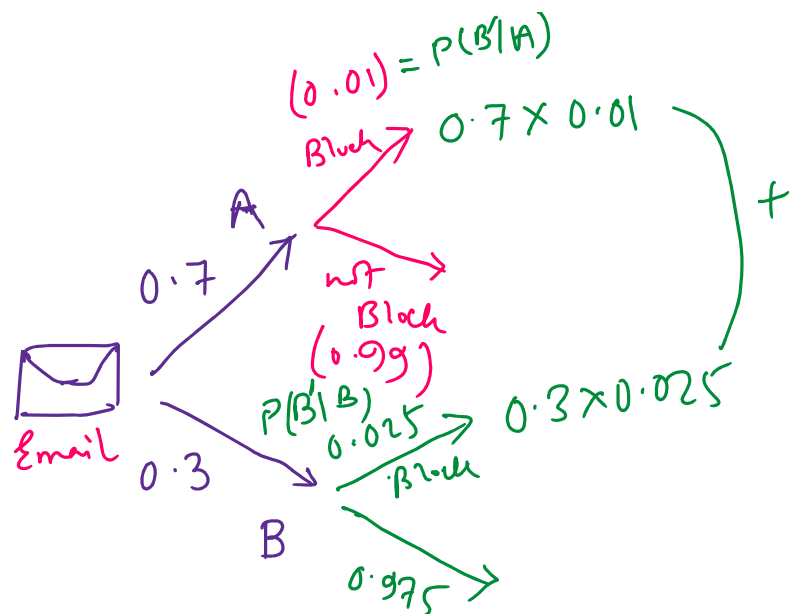
$P(A_1 \cap A_2)$
 $= P(A_2) \cdot P(A_1 | A_2)$

$P(A) = ? = P[(A \cap B_1) \cup (A \cap B_2) \cup (A \cap B_3) \cup (A \cap B_4)]$
 $= P(A \cap B_1) + P(A \cap B_2) + P(A \cap B_3) + P(A \cap B_4)$

$$P(A) = P(B_1) \cdot P(A|B_1) + P(B_2) \cdot P(A|B_2) + P(B_3) \cdot P(A|B_3) + P(B_4) \cdot P(A|B_4)$$

$$P(A) = \sum_{i=1}^n P(B_i) \cdot P(A|B_i)$$

Total Probability



$$P(\text{Block}) = 0.7 \times 0.01 + 0.3 \times 0.025$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

$$\Rightarrow P(B|A) \cdot P(A) = P(A \cap B)$$

$$\Rightarrow \frac{P(B|A) \cdot P(A)}{P(B)} = \frac{P(A \cap B)}{P(B)}$$

$$\Rightarrow \frac{P(A \cap B)}{P(B)} = \frac{P(B|A) \cdot P(A)}{P(B)}$$

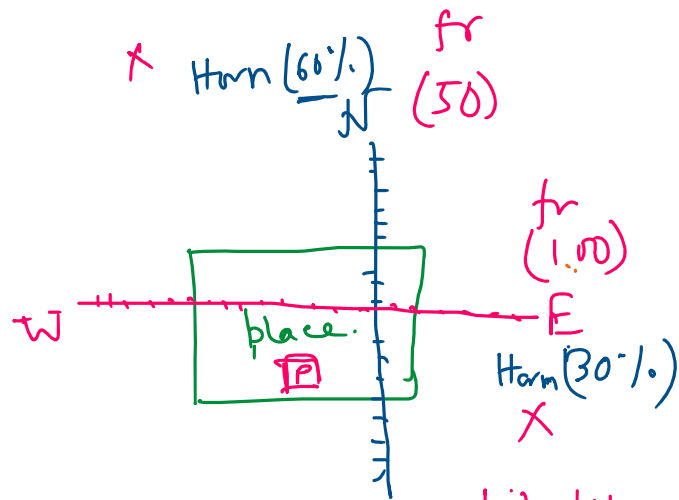
$$\Rightarrow \boxed{P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}}$$

Bayes' theorem.

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

$$A^c = \neg A$$

$$\begin{aligned} P(B) &= P(B|A) \cdot P(A) \\ &\quad + P(B|A^c) \cdot P(A^c) \\ &= P(B|A) \cdot P(A) + P(B|\neg A) \cdot P(\neg A) \end{aligned}$$



$$P(H|EW) = 0.3$$

$$P(H|NS) = 0.6$$

$$P(EW) = \frac{2}{3}$$

$$P(NS) = \frac{1}{3}$$

$$\begin{aligned} \therefore P(H) &= \frac{2}{3} \times 0.3 + \frac{1}{3} \times 0.6 \\ &= 0.2 + 0.2 = 0.4 \end{aligned}$$

$$\underbrace{P(EW|H)}_{\text{Posterior}} = \underbrace{\frac{P(H|EW)}{P(H)}}_{\text{Total Prob.}} \cdot \underbrace{P(EW)}_{\text{Prior}} = \frac{0.3}{0.4} \times \frac{2}{3} = \frac{1}{2} \quad (50\%)$$

$$P(L|N) = \frac{P(N|L) \cdot P(L)}{P(N)}$$

$$= \frac{P(N|L) \cdot P(L)}{P(N|L) \cdot P(L) + \underbrace{P(N|\neg L)}_{0.1} \cdot \underbrace{P(\neg L)}_{0.9}} = \frac{0.6 \times 0.1}{0.6 \times 0.1 + 0.1 \times 0.9}$$

$$= 0.3$$

$$P(A)$$

A

$$P(D|A)$$

$$= 5\%$$

$$P(B) = 0.7$$

B

$$P(D|B)$$

$$= 1\%$$

$$\therefore P(A|D) = \frac{P(D|A) \cdot P(A)}{P(D)}$$

$$= \frac{P(D|A) \cdot P(A)}{P(D|A) \cdot P(A) + P(D|B) \cdot P(B)}$$

$$= \frac{0.05 \times 0.3}{0.05 \times 0.3 + 0.01 \times 0.7} = 0.68$$

Problem: A credit card fraud system detects 100% of real frauds, but also flags 1% of legitimate transactions as fraud (false alarm). Real fraud is rare: 0.1% of all transactions. If the system flags a transaction, what is the probability it is actually fraud?

$$P(D | \text{Fraud}) = 1$$

$$P(D | F^c) = 0.01$$

$$P(F) = 0.1\% \\ = 0.001$$

$$P(F^c) = 0.999$$

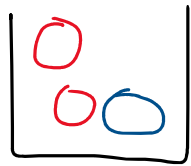
$$P(F | D) = \frac{P(D | F) \cdot P(F)}{P(D)}$$

$$= \frac{P(D | F) \cdot P(F)}{P(D | F) \cdot P(F) + P(D | F^c) \cdot P(F^c)}$$

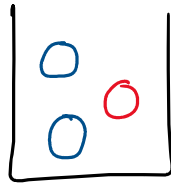
$$= \frac{1 \times 0.001}{1 \times 0.001 + 0.01 \times 0.999}$$

$$= \frac{0.001}{0.001 + 0.00999}$$

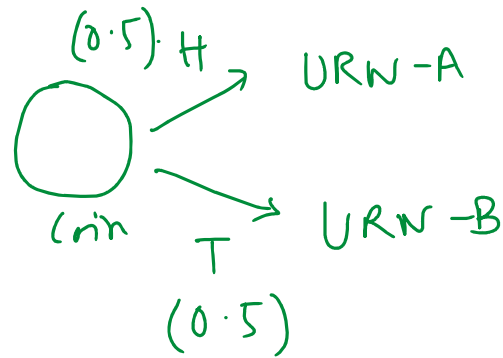
$$= \frac{0.001}{0.01099}$$



URN-A



URN-B



$$P(U_A) = 0.5$$

$$P(U_B) = 0.5$$

$$P(R|U_A) = \frac{2}{3}$$

$$P(R|U_B) = \frac{1}{3}$$

$$\underbrace{P(U_A|R)}_{\frac{2}{3}} =$$

$$\frac{P(R|U_A) \cdot P(U_A) \xrightarrow{\frac{2}{3}}}{\underbrace{P(R)}} = \frac{\frac{2}{3} \times \frac{1}{2}}{\frac{2}{3} \times \frac{1}{2} + \frac{1}{3} \times \frac{1}{2}} = \frac{\frac{2}{3}}{\frac{2}{3} + \frac{1}{3}}$$

$$P(R|U_A) \cdot P(U_A) + P(R|U_B) \cdot P(U_B) = \frac{2}{3}$$